在升級之前、如果沒有升級顧問、應該先確認哪些事項？

ONTAP 9

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在升級之前、如果沒有升級顧問、應該先確認哪些事項？

升級前的驗證事項

如果您不使用 "Active IQ" 升級顧問若要規劃升級、您應該在升級之前先確認叢集升級限制和叢集活動。

驗證叢集升級限制

如果您不使用 "Active IQ" 升級顧問、您必須確認叢集未超過平台系統限制。除了平台系統限制之外、SAN也有您應該驗證的限制。

1. 確認叢集未超過平台的系統限制。

   "NetApp Hardware Universe"

2. 如果您的叢集已設定SAN、請確認其未超過FC、FCoE和iSCSI的組態限制。

   "NetApp Hardware Universe"

3. 確定CPU和磁碟使用率：「節點執行節點node_name -command Sysstat -c 10 -x 3s」

   您應該監控CPU和磁碟使用率30秒。在所有報告的10項測量中、「* CPU*」和「磁碟使用率」欄中的值不應超過50%。在升級完成之前、不應將額外的負載新增至叢集。附註：CPU與磁碟使用率可能會因環境的不同時間而有所差異。因此、最好在預期的升級期間檢查CPU和磁碟使用率。

驗證目前的叢集活動

如果您不使用 "Active IQ" 升級顧問程式、在升級之前、您應該手動確認沒有任何工作正在執行、而且任何無法持續使用的CIFS工作階段都會終止。

確認沒有工作正在執行

在升級ONTAP 此功能之前、您必須先確認叢集工作的狀態。如果有任何Aggregate、Volume、NDMP（傾印或還原）或Snapshot工作（例如建立、刪除、移動、修改、複寫、 和掛載工作）正在執行或排入佇列、您必須允許工作成功完成或停止佇列的項目。

1. 檢閱任何執行中或佇列的Aggregate、Volume或Snapshot工作清單：「jobshow」

```
cluster1::> job show
                            Owning
Job ID Name                 Vserver    Node           State
------ -------------------- ---------- -------------- ----------
8629   Vol Reaper           cluster1   -              Queued
       Description: Vol Reaper Job
8630   Certificate Expiry Check
                            cluster1   -              Queued
       Description: Certificate Expiry Check
.
.
.
```

2. 如果有任何執行中的工作、請允許他們成功完成。
3. 刪除任何佇列的Aggregate、Volume或Snapshot複本工作：「jobdelete -id job_id」

```
cluster1::> job delete -id 8629
```

4. 確認沒有正在執行或排入佇列的Aggregate、Volume或Snapshot工作：「jobshow」

   在此範例中、所有執行中和佇列中的工作都已刪除：

```
cluster1::> job show
                            Owning
Job ID Name                 Vserver    Node           State
------ -------------------- ---------- -------------- ----------
9944   SnapMirrorDaemon_7_2147484678
                            cluster1   node1          Dormant
       Description: Snapmirror Daemon for 7_2147484678
18377  SnapMirror Service Job
                            cluster1   node0          Dormant
       Description: SnapMirror Service Job
2 entries were displayed
```

識別應終止的作用中**CIFS**工作階段

在升級ONTAP 完這個功能之前、您應該先識別並順利終止任何無法持續使用的CIFS工作階段。

持續可用的CIFS共用（可由Hyper-V或使用SMB 3.0傳輸協定的Microsoft SQL Server用戶端存取）不需要在升級之前終止。

1. 識別任何無法持續使用的已建立CIFS工作階段：「vserver CIFS工作階段顯示-持續可用的是執行個體」

此命令會顯示任何未持續可用的CIFS工作階段的詳細資訊。在繼續ONTAP 執行升級之前、您應該先終止這些更新。

```
cluster1::> vserver cifs session show -continuously-available Yes
-instance

                        Node: node1
                     Vserver: vs1
                  Session ID: 1
                Connection ID: 4160072788
Incoming Data LIF IP Address: 198.51.100.5
      Workstation IP address: 203.0.113.20
    Authentication Mechanism: NTLMv2
                Windows User: CIFSLAB\user1
                   UNIX User: nobody
                 Open Shares: 1
                  Open Files: 2
                  Open Other: 0
              Connected Time: 8m 39s
                   Idle Time: 7m 45s
            Protocol Version: SMB2_1
      Continuously Available: No
1 entry was displayed.
```

2. 如有必要、請識別您所識別的每個CIFS工作階段所開啟的檔案：「vserver CIFS工作階段檔案show -session-id session_ID」

```
cluster1::> vserver cifs session file show -session-id 1

Node:       node1
Vserver:    vs1
Connection: 4160072788
Session:    1
File    File      Open Hosting
Continuously
ID      Type      Mode Volume          Share                 Available
------- --------- ---- --------------- ---------------------
------------
1       Regular   rw   vol10           homedirshare          No
Path: \TestDocument.docx
2       Regular   rw   vol10           homedirshare          No
Path: \file1.txt
2 entries were displayed.
```

韌體ONTAP 如何在升級過程中更新

由於ONTAP 升級功能包括升級韌體、因此您不需要手動更新韌體。當您執行ONTAP 還原升級時、ONTAP 會將包含在升級套件中的叢集韌體複製到每個節點的開機裝置、並自動安裝新的韌體。

如果叢集中的韌體版本比ONTAP 隨附於此版更新套件的韌體版本舊、則會自動更新下列元件的韌體：

- BIOS /載入程式
- 服務處理器（SP）或基板管理控制器（BMC）
- 儲存櫃
- 磁碟
- Flash 快取

如有需要、您也可以在ONTAP 各個版本之間手動更新韌體。

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